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# Sharpness-Aware Minimization with Adaptive Regularization for Training Deep Neural Networks

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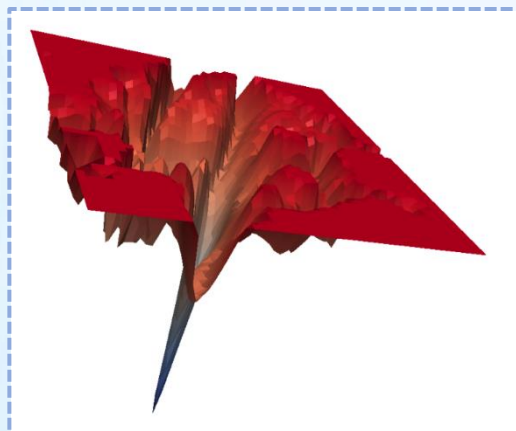
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# Background: Sharpness-Aware Minimization (SAM)\*

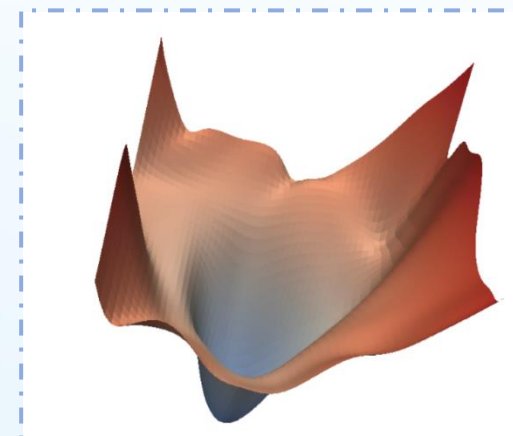


ResNet trained with SGD\*

Sharp minima VS Flat minima



**Flat minima can generalize better.**



ResNet trained with SAM\*

Loss landscape training with SGD and SAM

\*Sharpness-Aware Minimization for Efficiently Improving Generalization. ICLR, 2021.

# Comparison of Minimization Objective Function

(Stochastic Gradient Descent)

SGD

To solve  $\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$ ,

constructing numerical iteration:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k).$$

Only minimize training loss.

$-\nabla f(\mathbf{x}_k)$ : the *steepest descent direction* about loss.

(Sharpness-Aware Minimization)

SAM

To solve  $\min_{\mathbf{x} \in \mathbb{R}^d} \max_{\|\boldsymbol{\epsilon}\| \leq \rho} f(\mathbf{x} + \boldsymbol{\epsilon})$ ,

constructing numerical iteration:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k + \boldsymbol{\epsilon}_k),$$

$$\boldsymbol{\epsilon}_k = \rho \frac{\nabla f(\mathbf{x}_k)}{\|\nabla f(\mathbf{x}_k)\|}.$$

Minimize the maximal loss within neighborhood.

$-\nabla f(\mathbf{x}_k + \boldsymbol{\epsilon}_k)$ : the *direction generalizes better*.

# Rethink the Objective Function of SAM

Simple rewriting

$$\min_{\mathbf{x} \in \mathbb{R}^d} \max_{\|\boldsymbol{\epsilon}\| \leq \rho} f(\mathbf{x} + \boldsymbol{\epsilon}) = \min_{\mathbf{x} \in \mathbb{R}^d} \left[ 1 \cdot f(\mathbf{x}) + 1 \cdot \max_{\|\boldsymbol{\epsilon}\| \leq \rho} [f(\mathbf{x} + \boldsymbol{\epsilon}) - f(\mathbf{x})] \right].$$

The weight ratio is **fixed** between loss term and sharpness term (1:1).

Potential issues:

- **Limited flexibility**
- **Deficient dynamics**

# Our Method

*Can we reach reasonable trade-off between loss reduction and generalization enhancement?*



Introducing **adaptive weight** into the objective function:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \left[ 1 \cdot f(\mathbf{x}) + \lambda_k \cdot \max_{\|\boldsymbol{\epsilon}\| \leq \rho} [f(\mathbf{x} + \boldsymbol{\epsilon}) - f(\mathbf{x})] \right].$$

Construct the numerical iteration:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{x}_k - \eta (\lambda_k \nabla f(\mathbf{x}_k + \boldsymbol{\epsilon}_k) + (1 - \lambda_k) \nabla f(\mathbf{x}_k)), \\ \boldsymbol{\epsilon}_k &= \rho \frac{\nabla f(\mathbf{x}_k)}{\|\nabla f(\mathbf{x}_k)\|}. \end{aligned}$$

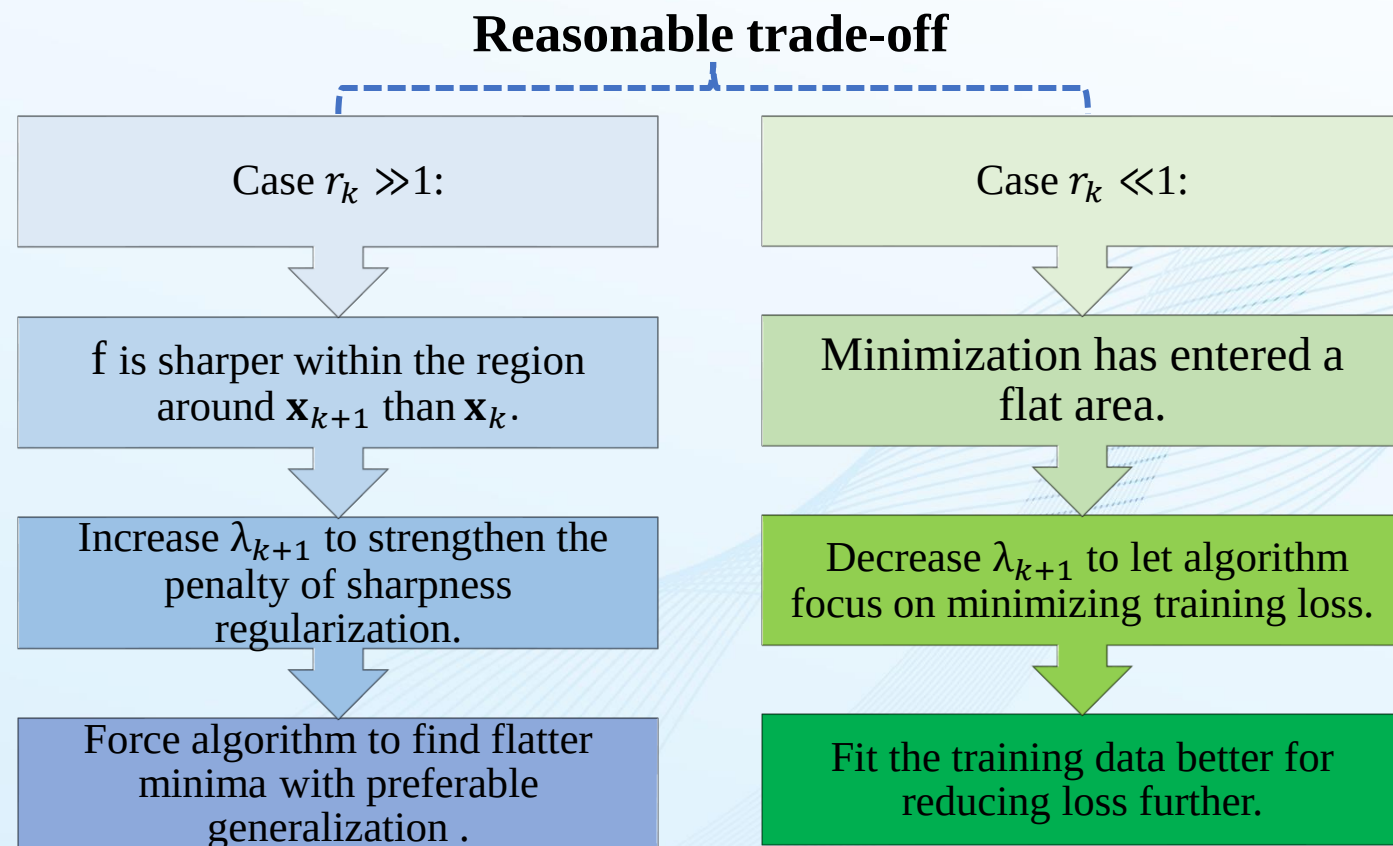
# Adaptive Strategy

*How to update the adaptive weight of sharpness term to achieve rational balance?*

Design the following **sharpness ratio rule**:

$$\begin{aligned}
 r_k &\triangleq \frac{|f(\mathbf{x}_k + \boldsymbol{\epsilon}_k) - f(\mathbf{x}_k)|}{|f(\mathbf{x}_{k-1} + \boldsymbol{\epsilon}_{k-1}) - f(\mathbf{x}_{k-1})|} \\
 &\approx \frac{|\langle \nabla f(\mathbf{x}_k), \boldsymbol{\epsilon}_k \rangle|}{|\langle \nabla f(\mathbf{x}_{k-1}), \boldsymbol{\epsilon}_{k-1} \rangle|} \\
 &= \frac{\|\nabla f(\mathbf{x}_k)\|}{\|\nabla f(\mathbf{x}_{k-1})\|}
 \end{aligned}$$

**if**  $r_k \geq \chi$ :  $\lambda_{k+1} = \mathbf{Proj}_{[\delta, 1-\delta]}(\gamma \lambda_k)$ ,  
**else**:  $\lambda_{k+1} = \mathbf{Proj}_{[\delta, 1-\delta]}(\frac{1}{\gamma} \lambda_k)$ ,



# Theoretical Property

The iterable sequence  $\{\mathbf{x}_k\}$  generated by SAMAR converges at rate of  $\mathcal{O}(1/\sqrt{K})$ .

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{x}_k)\|^2] \leq \mathcal{O}(1/\sqrt{K}).$$

**Algorithm 1** Sharpness Aware Minimization with Adaptive Regularization

**Initialize:**  $\mathbf{x}_0, \rho, \eta_0, 0 < \lambda_0 < 1, \chi, \gamma > 1, 0 < \delta < 1$ .

**for**  $k = 0, \dots, K - 1$  **do**

**step 1:** Compute the stochastic gradient  $\mathbf{g}(\mathbf{x}_k)$  on the batch data.

**step 2:** Calculate  $\epsilon_{\mathbf{x}_k} = \rho \frac{\mathbf{g}(\mathbf{x}_k)}{\|\mathbf{g}(\mathbf{x}_k)\|}$  through first-order stochastic linearization approximation.

**step 3:** Let  $\mathbf{s}_k = (1 - \lambda_k)\mathbf{g}(\mathbf{x}_k) + \lambda_k\mathbf{g}(\mathbf{x}_k + \epsilon_{\mathbf{x}_k})$ .

**step 4:**  $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \mathbf{s}_k$ .

**step 5:** Calculate  $r_k = \frac{\|\mathbf{g}(\mathbf{x}_k)\|}{\|\mathbf{g}(\mathbf{x}_{k-1})\|}$ .

**step 6:**

**if**  $r_k \geq \chi$  or  $k = 0$  :  $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\gamma \lambda_k)$ .

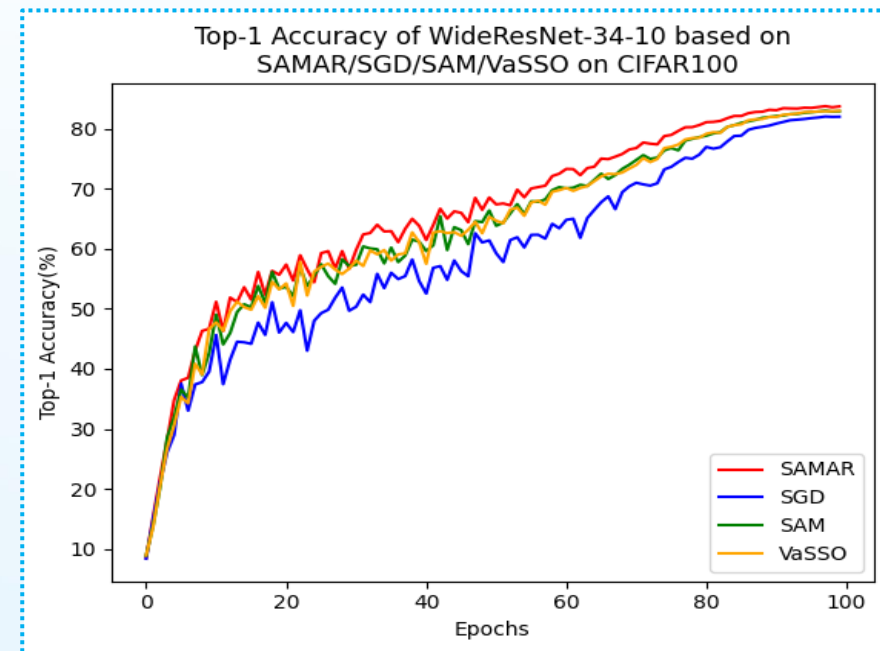
**else :**  $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\frac{1}{\gamma} \lambda_k)$ .

**step 7:**  $k = k + 1$ .

**end for**

**return**  $\mathbf{x}_K$

# Numerical Results



We train ResNet-34-10 neural network on CIFAR-100 dataset using **SAMAR/SGD/SAM/VaSSO** optimizers and record the top-1 accuracy. Training curvatures demonstrate that SAMAR outperforms other rivals.

# Conclusions

- We propose a SAM-type method with adaptive regularization weight to overcome the fixed weight ratio shortage.
- Theoretically, SAMAR converges at rate of  $O(1/\sqrt{K})$ .
- Image classification task showcases that SAMAR attains relatively notable improvement compared with several optimizers.

## Future Work

- Test the performance of SAMAR on more complex **ImageNet** dataset
- Extend to other deep learning tasks, liking **machine translation, transfer learning**, etc.

# Thank You

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